

Patient

Sex: Male
Case Number: TN24-276125
Diagnosis: Neoplasm, NOS

Specimen Information

Primary Tumor Site: Upper lobe, lung
Specimen Site: Upper lobe, lung
Specimen ID: S24-30778-A4
Specimen Collected: 01-Nov-2024
Test Report Date: 11-Dec-2024

Ordered By

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Results with Therapy Associations

BIOMARKER	METHOD	ANALYTE	RESULT	THERAPY ASSOCIATION		BIOMARKER LEVEL*
PD-L1 (22c3)	IHC	Protein	Positive, TPS: 1%	BENEFIT	pembrolizumab	Level 1
PD-L1 (28-8)	IHC	Protein	Positive 1+, 5%	BENEFIT	nivolumab/ipilimumab combination	Level 1
KRAS	Seq	DNA-Tumor	Pathogenic Variant Exon 2 p.G12C	BENEFIT	adagrasib, sotorasib	Level 2
ALK	IHC	Protein	Negative 0	LACK OF BENEFIT	alectinib, ceritinib, crizotinib, lorlatinib	Level 1
					brigatinib	Level 2
	Seq	RNA-Tumor	Fusion Not Detected		alectinib, brigatinib, ceritinib, crizotinib, lorlatinib	Level 2
BRAF	Seq	DNA-Tumor	Mutation Not Detected	LACK OF BENEFIT	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	Level 2
EGFR	Seq	DNA-Tumor	Mutation Not Detected	LACK OF BENEFIT	afatinib, dacomitinib, erlotinib, gefitinib, osimertinib	Level 2
MET	Seq	RNA-Tumor	Variant Transcript Not Detected	LACK OF BENEFIT	capmatinib, tepotinib	Level 2
					crizotinib	Level 2
RET	Seq	RNA-Tumor	Fusion Not Detected	LACK OF BENEFIT	pralsetinib, selpercatinib	Level 2
ROS1	Seq	RNA-Tumor	Fusion Not Detected	LACK OF BENEFIT	ceritinib, crizotinib, entrectinib, lorlatinib, repretrectinib	Level 2

* Biomarker reporting classification: Level 1 Companion diagnostic (CDx); Level 2 Strong evidence of clinical significance or is endorsed by standard clinical guidelines; Level 3 Potential clinical significance. Bolded benefit therapies, if present, highlight the most clinically significant findings.

The selection of any, all, or none of the matched therapies resides solely with the discretion of the treating physician. Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all available information concerning the patient's condition, the FDA prescribing information for any therapeutic, and in accordance with the applicable standard of care. Whether or not a particular patient will benefit from a selected therapy is based on many factors and can vary significantly. All trademarks and registered trademarks are the property of their respective owners.

Important Note

This patient has a potential NCI-ComboMATCH Trial-eligible result. Please see the clinical trials section on *Page 6*.

For PD-L1 TPS between 1-49%, the NCCN guidelines recommend pembrolizumab plus chemotherapy: cisplatin or carboplatin + pemetrexed for nonsquamous histologies and carboplatin + paclitaxel or nab-paclitaxel for squamous cell carcinoma.

Please note that multiple companion diagnostic assays (antibodies) have been utilized to assess PD-L1 expression. Each test has different performance characteristics, therefore, the results will not always be concordant.

For information on Caris GPSai™, please refer to relevant section of the report (Additional Results Page 1).

Cancer-Type Relevant Biomarkers

Biomarker	Method	Analyte	Result	Biomarker	Method	Analyte	Result
MET	Seq	RNA-Tumor	Likely Pathogenic Fusion	FGFR3	Seq	RNA-Tumor	Fusion Not Detected
	CNA-Seq	DNA-Tumor	Amplification Not Detected	KEAP1	Seq	DNA-Tumor	Variant of Uncertain Significance Exon 6 p.V608fs
	IHC	Protein	Negative 1+, 20%	MTAP	CNA-Seq	DNA-Tumor	Deletion Not Detected
	Seq	DNA-Tumor	Mutation Not Detected	NFE2L2	Seq	DNA-Tumor	Mutation Not Detected
STK11	Seq	DNA-Tumor	Likely Pathogenic Variant Exon 4 p.P179L	NRG1	Seq	RNA-Tumor	Fusion Not Detected
MSI	Seq	DNA-Tumor	Stable	PD-L1 (SP142)	IHC	Protein	Negative, IC: 1% Negative, TC: 0%
NTRK1/2/3	Seq	RNA-Tumor	Fusion Not Detected	PD-L1 (SP263)	IHC	Protein	Negative, TC: 0%
Tumor Mutational Burden	Seq	DNA-Tumor	Low, 8 mut/Mb	RB1	Seq	DNA-Tumor	Mutation Not Detected
ALK	Seq	DNA-Tumor	Mutation Not Detected	RET	Seq	DNA-Tumor	Mutation Not Detected
BRAF	Seq	RNA-Tumor	Fusion Not Detected	TP53	Seq	DNA-Tumor	Mutation Not Detected
ERBB2 (Her2/Neu)	CNA-Seq	DNA-Tumor	Amplification Not Detected				
	IHC	Protein	Negative Score 0				
	Seq	DNA-Tumor	Mutation Not Detected				

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Genomic Signatures

Biomarker	Method	Analyte	Result
Microsatellite Instability (MSI)	Seq	DNA-Tumor	Stable
Tumor Mutational Burden (TMB)	Seq	DNA-Tumor	Result: Low 8 Low10High
Genomic Loss of Heterozygosity (LOH)	Seq	DNA-Tumor	Low - 5% of tested genomic segments exhibited LOH (assay threshold is ≥ 16%)

Genes Tested with Pathogenic or Likely Pathogenic Alterations

Gene	Method	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %
ARID1A	Seq	DNA-Tumor	Pathogenic Variant	p.Y1260fs	15	c.3777dupC	9
CDKN2A	Seq	DNA-Tumor	Likely Pathogenic Variant	p.V28_E33del	1	c.83_100del18	23
	Seq	DNA-Tumor	Pathogenic Variant	p.V126D	2	c.377T>A	12
KRAS	Seq	DNA-Tumor	Pathogenic Variant	p.G12C	2	c.34G>T	43
MET	Seq	RNA-Tumor	Likely Pathogenic Fusion	SND1-MET	6	-	-
STK11	Seq	DNA-Tumor	Likely Pathogenic Variant	p.P179L	4	c.536C>T	23

Unclassified alterations for DNA and RNA sequencing can be found in the MI Portal.
Formal nucleotide nomenclature and gene reference sequences can be found in the Appendix of this report.

Genes Tested with Variants of Uncertain Significance

Gene	Method	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %
KEAP1	Seq	DNA-Tumor	Variant of Uncertain Significance	p.V608fs	6	c.1821dupT	19

Additional Variants of Uncertain Significance can be found in the MI Portal.

Human Leukocyte Antigen (HLA) Genotype Results

The impact of HLA genotypes on drug response and prognosis is an active area of research. These results can help direct patients to clinical trials recruiting for specific genotypes. Please see www.clinicaltrials.gov for more information.

Gene	Method	Analyte	Genotype
MHC CLASS I			
HLA-A	Seq	DNA-Tumor	A*24:02, A*24:02
HLA-B	Seq	DNA-Tumor	B*15:01, B*55:01
HLA-C	Seq	DNA-Tumor	C*03:03, C*03:03

HLA genotypes with only one allele are either homozygous or have loss of heterozygosity at that position.

Immunohistochemistry Results

Biomarker	Result	Biomarker	Result
ALK	Negative 0	PD-L1 (28-8)	Positive 1+, 5%
ERBB2 (Her2/Neu)	Negative Score 0	PD-L1 (SP142)	Negative, IC: 1% Negative, TC: 0%
MET	Negative 1+, 20%	PD-L1 (SP263)	Negative, TC: 0%
PD-L1 (22c3)	Positive, TPS: 1%		

Genes Tested with Indeterminate Results by Tumor DNA Sequencing

COL2A1	SMARCA2										
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Genes in this table were ruled indeterminate due to low coverage for some or all exons.

The results in this report were curated to represent biomarkers most relevant for the submitted cancer type. These include results important for therapeutic decision making, as well as notable alterations in other biomarkers known to be involved in oncogenesis. Additional results, including genes with normal findings, variants of uncertain significance, or unclassified alterations can be found in the MI Portal at miportal.carismolecularintelligence.com. If you do not have an MI Portal account, or need assistance accessing it, please contact Caris Customer Support at (888) 979 8669.

Notes of Significance

SEE APPENDIX FOR DETAILS

Clinical Trials Connector[™] opportunities based on biomarker expression: 632 Targeted Therapies. See page 6 for details.

Specimen Information

Specimen ID: S24 30778 A4

Specimen Collected: 11/01/2024

Specimen Received: 11/27/2024

Testing Initiated: 11/30/2024

Test Ordered*: MI Profile[™] (MI Tumor Seek Hybrid[™] + IHCs and Other Tests by Tumor Type)

* If the submitted specimen is inadequate, only a subset of the ordered testing may be reported.

Gross Description: 1 (A) Paraffin Block Client ID(S24 30778 A4) from UK HealthCare Albert B Chandler Hospital, Lexington, KY, with the corresponding pathology report labeled "S24 30778".

Dissection Information: Molecular testing of this specimen was performed after harvesting of targeted tissues with an approved manual microdissection technique. Candidate slides were examined under a microscope and areas containing tumor cells (and separately normal cells, when necessary for testing) were circled. A laboratory technician harvested targeted tissues for extraction from the marked areas using a dissection microscope.

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Clinical Trials Connector™

The Clinical Trials Connector lists agents that are matched to available clinical trials according to biomarker status. In some instances, older generation agents may still be relevant in the context of new combination strategies and, therefore, will still appear on this report.

Therapeutic agents listed below may or may not be currently FDA approved for the tumor type tested.

Please see <https://clinicaltrials.gov/> for more information.

NCI-COMBOMATCH BIOMARKER SUMMARY

Description	Biomarker	Method	Analyte	Investigational Agent(s)
KRAS G12C mutation / sotorasib + panitumumab	KRAS	Seq	DNA-Tumor	sotorasib + panitumumab

The patient may be eligible to enroll in a treatment study under NCI ComboMATCH. Please note, this does not guarantee enrollment to a treatment study, as some treatment trials have additional inclusion criteria. Further evaluation by the ComboMATCH protocol team will be necessary. Please contact NCI for confirmation.

TARGETED THERAPIES (632)

Drug Class	Biomarker	Method	Analyte	Investigational Agent(s)
ATR inhibitors (8)	ARID1A	NGS	DNA-Tumor	berzosertib, M1774, RP-3500
BET bromodomain inhibitors (4)	ARID1A	NGS	DNA-Tumor	CPI-0610, tazemetostat
Cell cycle inhibitors (24)	CDKN2A	NGS	DNA-Tumor	abemaciclib, BLU-222, palbociclib, PF-07220060, ribociclib
cMET-targeted therapy (51)	MET	RNA-Seq	RNA-Tumor	amivantamab, APL-101, bozitinib, brigatinib, cabozantinib, capmatinib, crizotinib, MYTX-011, savolitinib, telisotuzumab vedotin, tepotinib
Dual MEK/RAF inhibitor (2)	KRAS	NGS	DNA-Tumor	avutometinib
Heat shock pathway inhibitors (1)	ARID1A	NGS	DNA-Tumor	NXP800
Immunomodulatory agents (409)	PD-L1	IHC	Protein	atezolizumab, avelumab, balstilimab, cemiplimab, cetrelimab, dostarlimab, durvalumab, envafohimab, ipilimumab, nivolumab, pembrolizumab, retifanlimab, spartalizumab, tislelizumab, toripalimab, tremelimumab, XmAb20717, zimberelimab
KRAS G12C inhibitors (29)	KRAS	NGS	DNA-Tumor	adagrasib, BI 1701963, D-1553, GDC-6036, JAB-21822, sotorasib
MEK inhibitors (25)	KRAS	NGS	DNA-Tumor	binimetinib, cobimetinib, mirdametinib, trametinib
PARP inhibitors (28)	ARID1A	NGS	DNA-Tumor	BGB-290, niraparib, olaparib, pamiparib, senaparib, talazoparib
PI3K/Akt/mTOR inhibitors (39)	STK11	NGS	DNA-Tumor	gedatolisib, paxalisib, everolimus, rapamycin, nab-sirolimus, sirolimus, temsirolimus

() = represents the total number of clinical trials identified by the Clinical Trials Connector for the provided drug class or table.

The Clinical Trials Connector may include trials that enroll patients with additional screening of molecular alterations. In some instances, only specific gene variants may be eligible.

Additional Clinical Trials Connector results continued on the next page. >

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Clinical Trials Connector[™]

TARGETED THERAPIES (632)				
Drug Class	Biomarker	Method	Analyte	Investigational Agent(s)
SHP2 inhibitors (10)	KRAS	NGS	DNA-Tumor	BBP-398, HBI-2376, JAB-3312, RLY-1971, RMC-4630, TNO155
SWI/SNF complex inhibitors (2)	ARID1A	NGS	DNA-Tumor	PRT3789

() = represents the total number of clinical trials identified by the Clinical Trials Connector for the provided drug class or table.

The Clinical Trials Connector may include trials that enroll patients with additional screening of molecular alterations. In some instances, only specific gene variants may be eligible.

Disclaimer

Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all available information concerning the patient's condition, prescribing information for any therapeutic, and in accordance with the applicable standard of care. Drug associations provided in this report do not guarantee that any particular agent will be effective for the treatment of any patient or for any particular condition. Caris Life Sciences[®] expressly disclaims and makes no representation or warranty whatsoever relating, directly or indirectly, to the performance of services, including any information provided and/or conclusions drawn from therapies that are included or omitted from this report. Whether or not a particular patient will benefit from a selected therapy is based on many factors and can vary significantly. The selection of therapy, if any, resides solely in the discretion of the treating physician and the tests should not be considered a companion diagnostic.

Caris MPI, Inc. d/b/a Caris Life Sciences is certified under the Clinical Laboratory Improvement Amendments (CLIA) as qualified to perform high complexity clinical laboratory testing, including all Caris molecular profiling assays. Individual assays that are available through Caris molecular profiling include both Laboratory Developed Tests (LDT) and U.S. Food and Drug Administration (FDA) approved or cleared tests. In addition, certain tests have been CE marked as a general IVD under the In Vitro Diagnostic Directive (IVDD) 98/79/EC. Offered LDTs were developed and their performance characteristics determined by Caris. Certain tests have not been cleared or approved by the FDA. Caris LDTs are used for clinical purposes. They are not investigational or for research. Caris' CLIA certification number is located at the bottom of each page of this report.

The information presented in the Clinical Trials Connector[™] section of this report, if applicable, is compiled from sources believed to be reliable and current. However, the accuracy and completeness of the information provided herein cannot be guaranteed. The clinical trials information present in the biomarker description was compiled from www.clinicaltrials.gov. The contents are to be used only as a guide, and health care providers should employ their best comprehensive judgment in interpreting this information for a particular patient. Specific eligibility criteria for each clinical trial should be reviewed as additional inclusion criteria may apply.

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Caris molecular testing is subject to Caris' intellectual property. Patent www.CarisLifeSciences.com/ip.



Electronic Signature
Hassan Ghani, MD
11 Dec 2024



The Caris GPSai (Caris Genomic Probability Score – Artificial Intelligence) is a cancer-type similarity assessment which compares the characteristics of a patient's tumor against other tumors in the Caris database. Caris GPSai analyzes a tumor's molecular signature and provides the prevalence of that signature in the Caris Life Sciences genomic and transcriptomic database across 90 distinct cancer categories.

Most Likely Tumor Type: Lung Adenocarcinoma

90%

The predicted probability of each cancer type that can be detected by the product are shown below. The total probabilities of the top level cancer type (blue) total 100%. The subtypes (indented) sum to the total probability predicted for the top level cancer type. Discrepancies between expected sums and displayed sums may be exhibited due to rounding.

0	Adrenal Cortical Carcinoma
0	Bladder/Urinary Tract
0	Urothelial Carcinoma
0	Bladder Adenocarcinoma
0	Bowel
0	Appendiceal Adenocarcinoma
0	Colorectal Adenocarcinoma
0	Small Bowel Carcinoma
0	Breast
0	Metaplastic Breast Cancer
0	Breast Invasive Lobular Carcinoma
0	Breast Invasive Ductal Carcinoma
0	CNS/Brain
0	Diffuse Glioma
0	Meningioma
0	Cervix/Uterine Carcinoma
0	Uterine Serous Carcinoma/Uterine Papillary Serous Carcinoma
0	Uterine Carcinosarcoma/Uterine Malignant Mixed Mullerian Tumor
0	Uterine Clear Cell Carcinoma
0	Cervical Adenocarcinoma
0	Uterine Endometrioid Carcinoma
0	Cutaneous Squamous Cell Carcinoma
0	Esophagus/Stomach
0	Esophageal Squamous Cell Carcinoma
0	Esophagogastric Adenocarcinoma
0	Stomach Adenocarcinoma
0	Adenocarcinoma of the Gastroesophageal Junction
0	Esophageal Adenocarcinoma
0	Germ Cell Tumor
0	Hematological
0	Hepatocellular Carcinoma
0	Kidney
0	Renal Cell Carcinoma
0	Chromophobe Renal Cell Carcinoma
0	Papillary Renal Cell Carcinoma
0	Renal Clear Cell Carcinoma
0	Wilms Tumor
0	Melanoma
0	Mesothelioma
7	Neuroendocrine Neoplasm
4	Well/Moderately-Differentiated Neuroendocrine Tumor
3	Poorly-Differentiated Neuroendocrine Carcinoma
3	Large Cell Neuroendocrine Carcinoma
0	Small Cell Neuroendocrine Carcinoma
0	Paraganglioma/Pheochromocytoma
0	Merkel Cell Carcinoma

90	Non-Small Cell Lung Carcinoma
0	Lung Squamous Cell Carcinoma
90	Lung Adenocarcinoma
0	Orogenital Squamous Cell Carcinoma
0	Ovarian Epithelial Tumor
0	Ovarian Carcinosarcoma/Malignant Mixed Mesodermal Tumor
0	Serous Ovarian/Fallopian Tube/Peritoneal
0	Low-Grade Serous Ovarian/Fallopian Tube/Peritoneal Cancer
0	High-Grade Serous Ovarian/Fallopian Tube/Peritoneal Cancer
0	Mucinous Ovarian Cancer
0	Clear Cell Ovarian Cancer
0	Endometrioid Ovarian Cancer
1	Pancreatobiliary
0	Gallbladder Cancer
0	Pancreatic Adenocarcinoma
0	Cholangiocarcinoma
0	Peripheral Nervous System
0	Schwannoma
0	Neuroblastoma
0	Malignant Peripheral Nerve Sheath Tumor
0	Prostate Adenocarcinoma
0	Salivary Gland Tumor
0	Sex Cord Stromal Tumor
0	Granulosa Cell Tumor
0	Sertoli-Leydig Cell Tumor
0	Soft Tissue/Bone
0	Gastrointestinal Stromal Tumor
0	Leiomyosarcoma
0	Angiosarcoma
0	Rhabdomyosarcoma
0	Synovial Sarcoma
0	Osteosarcoma
0	Liposarcoma
0	Chondrosarcoma
0	Endometrial Stromal Sarcoma
0	Ewing Sarcoma
0	Thymic Carcinoma
0	Thyroid
0	Medullary Thyroid Cancer
0	Hurthle Cell Thyroid Cancer
0	Well-Differentiated Thyroid Cancer
0	Follicular Thyroid Cancer
0	Papillary Thyroid Cancer
0	Anaplastic Thyroid Cancer

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Gene Expression

Gene	Percentile in Cancer Type	Gene	Percentile in Cancer Type	Gene	Percentile in Cancer Type
ADORA2A	84	FOLR1	56	NTRK3	90
ALK	89	GPC3	44	PDCD1	86
ASCL1	98	HGF	64	PDCD1LG2	42
ATM	18	HRAS	82	PIK3CA	12
AURKA	60	IGF1R	55	POU3F2	84
BRAF	26	KDM1A	42	PRAME	36
BRCA1	58	KDR	48	PTEN	12
BRCA2	58	KEAP1	55	RB1	50
BRD4	52	KRAS	66	RET	98
CCND1	96	LAG3	94	ROR1	57
CCND2	90	MAGEA4	75	ROR2	80
CCNE1	55	MDM2	24	ROS1	72
CD274	66	MET	34	SLFN11	58
CD276	74	MKI67	24	SRC	39
CDKN2A	73	MSLN	56	SSTR2	94
CEACAM5	93	MTAP	32	SSTR3	94
CLDN18	82	MTOR	12	SSTR5	98
CLDN6	60	MUC1	50	STK11	44
CTLA4	92	MUC16	46	TACSTD2	11
DLL3	96	MYC	29	TGFB1	61
EGFR	16	NECTIN4	50	TOP1	79
EPHA2	54	NEUROD1	98	TP53	96
ERBB2	27	NF1	22	TSC1	14
ERBB3	91	NRAS	12	TSC2	86
FGFR1	70	NRG1	91	VEGFA	18
FGFR2	57	NTRK1	91	XPO1	2
FGFR3	58	NTRK2	70	YAP1	4

Gene Expression of Selected Genes by Whole Transcriptome Sequencing (WTS) Methods:

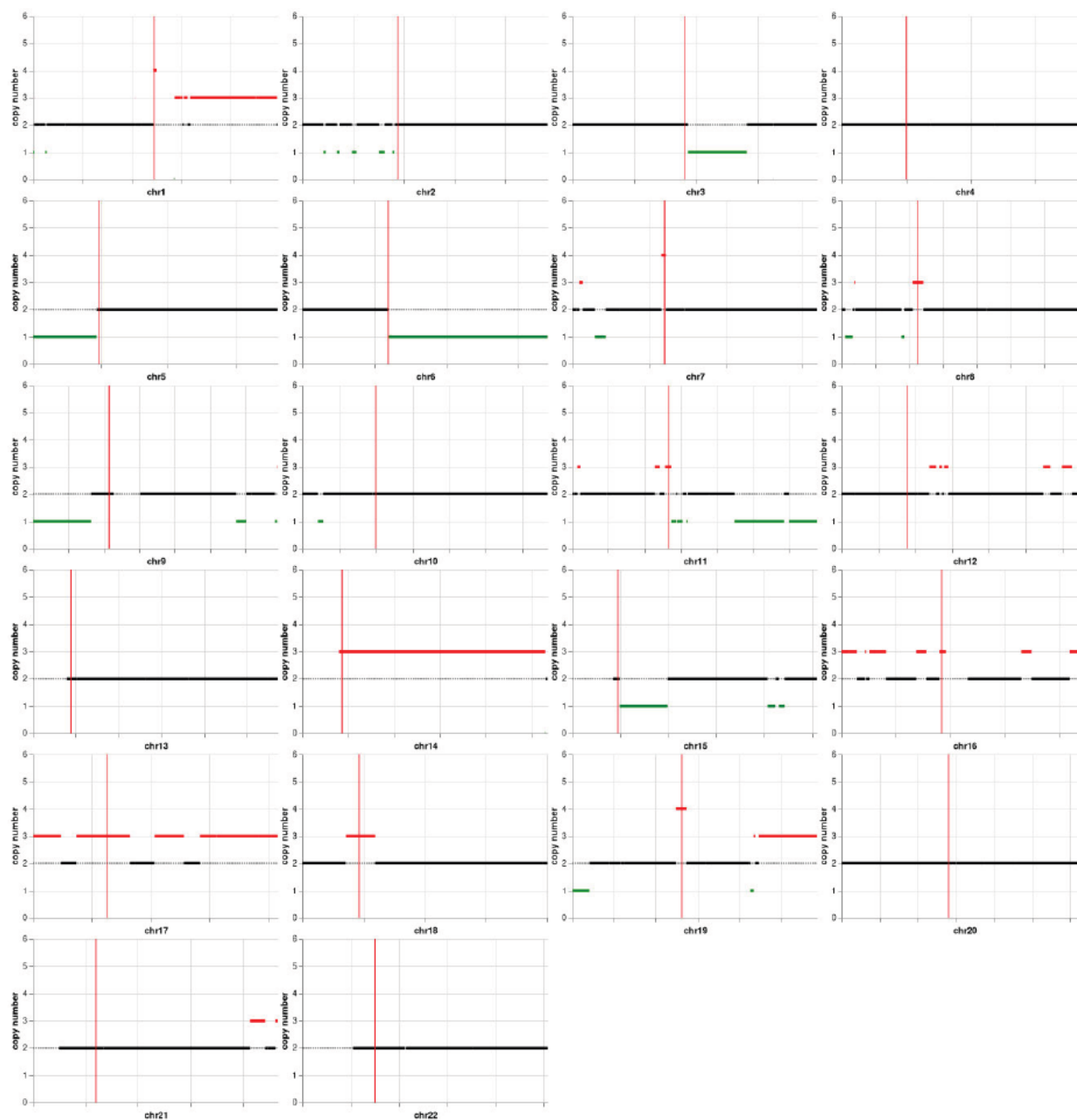
Gene expression is derived from whole transcriptome sequencing. Relative expression of genes are calculated as normalized values using transcripts per Million Molecules or TPM. TPM is presented as a percentile derived by comparison to a distribution of Caris' internal cohort of the tumor-type profiled. Selected genes reported in this section were chosen based on their tumor-type specific relevance for matching to clinical trials, or tumor type subclassification.

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Karyotype



Karyotyping using Copy Number Analysis by Whole Exome Sequencing (WES) Methods:

Whole exome sequencing in combination with interrogation of single nucleotide polymorphisms (SNPs) tiled throughout the genome, allows for the identification and visualization of cytogenetic aberrations.

Somatic structural variants like whole or partial chromosome duplications or deletions, are important for cancer development and progression, and may identify clinically actionable alterations.

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Mutational Analysis by Next-Generation Sequencing (NGS)

TUMOR MUTATIONAL BURDEN	
Mutations / Megabase	Result
8	Low

TMB

Tumor Mutational Burden (TMB) is defined as the number of somatic non synonymous mutations per million bases of sequenced DNA in a tumor sample. Tumors with high TMB may increase the number of neoantigens which is hypothesized to increase T cell reactivity and potential for response to immune checkpoint inhibitors. TMB analysis was performed based on next generation sequencing analysis of genomic DNA isolated from a tumor sample.

MICROSATELLITE INSTABILITY ANALYSIS	
Test	Result
MSI	Stable

MSI

Microsatellite instability (MSI) status is a measure of the number of somatic mutations within short, repeated sequences of DNA (microsatellites). MSI High status can indicate that the tumor has a defect in mismatch repair (MMR) abrogating the ability to correct mistakes during DNA replication. Tumors with MSI high status may increase the number of neoantigens which is hypothesized to increase T cell reactivity and potential for response to immune checkpoint inhibitors. Tumor only microsatellite instability status by NGS (MSI NGS) is measured by the direct analysis of known microsatellite regions sequenced in the CMI NGS panel.

GENOMIC LOSS OF HETEROZYGOSITY	
Test	Result
Genomic Loss of Heterozygosity (LOH)	Low - 5% of tested genomic segments exhibited LOH (assay threshold is $\geq 16\%$)

LOH

To calculate genomic loss of heterozygosity (LOH), the 22 autosomal chromosomes are split into 552 segments and the LOH of single nucleotide polymorphisms (SNPs) within each segment is calculated. Caris WES data consist of approximately 250k SNPs spread across the genome. SNP alleles with frequencies skewed towards 0 or 100% indicate LOH (heterozygous SNP alleles have a frequency of 50%). The final call of genomic LOH is based on the percentage of all 552 segments with observed LOH.

Additional Next-Generation Sequencing results continued on the next page. >

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Mutational Analysis by Next-Generation Sequencing (NGS)

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
ARID1A	DNA-Tumor	Pathogenic Variant	p.Y1260fs	15	c.3777dupC	9	NM_006015.5

Interpretation: A loss of function pathogenic frameshift mutation was found.

This gene encodes a member of the SWI/SNF family, whose members have helicase and ATPase activities and are thought to regulate transcription of certain genes by altering the chromatin structure around those genes. Inactivating mutations of ARID1A, a member of the SWI/SNF chromatin-remodeling complex, have been identified in a long list of cancers, including ovarian clear-cell carcinoma, gastric, hepatocellular, breast and so on. Mutational and functional data suggest ARID1A is a bona fide tumor suppressor. ARID1A may contribute to tumor suppression via effects on the SWI/SNF complex, control of cell proliferation and differentiation, and/or effects on histone ubiquitylation.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
CDKN2A	DNA-Tumor	Likely Pathogenic Variant	p.V28_E33del	1	c.83_100del18	23	NM_000077.4

Interpretation: This in frame CDKN2A deletion has been identified in a number of cancers, and it has not been reported as a germline variant. Thus, it is presumed to be pathogenic.

CDKN2A, or cyclin-dependent kinase inhibitor 2A, is a tumor suppressor gene that encodes two cell-cycle regulatory proteins, p16INK4A and p14ARF. As upstream regulators of the retinoblastoma (RB) and p53 signaling pathways, CDKN2A controls the induction of cell cycle arrest in damaged cells that allows for repair of DNA. Loss of CDKN2A through whole-gene deletion, point mutation, or promoter methylation leads to disruption of these regulatory proteins and consequently dysregulation of growth control. Somatic CDKN2A mutations are documented to occur in squamous cell lung cancers, head and neck cancer, colorectal cancer, chronic myelogenous leukemia and malignant pleural mesothelioma. Currently, there are agents that target downstream of CDKN2A such as CDK4/6 inhibitors which function by restoring the cell's ability to induce cell cycle arrest. Germline CDKN2A mutations are associated with melanoma-pancreatic carcinoma syndrome, which increases the risk for familial malignant melanoma and pancreatic cancer.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
CDKN2A	DNA-Tumor	Pathogenic Variant	p.V126D	2	c.377T>A	12	NM_000077.4

Interpretation: A pathogenic mutation was detected in CDKN2A (p16). This mutation (also known as c.377T>A) has been reported as a frequent germline mutation, causal for hereditary predisposition to melanoma (Goldstein 2001 Br J Cancer 85:527).

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
KEAP1	DNA-Tumor	Variant of Uncertain Significance	p.V608fs	6	c.1821dupT	19	NM_012289.3

Interpretation: A variant with no known clinical or functional significance was detected in KEAP1.

This gene encodes a protein containing KELCH-1 like domains, as well as a BTB/POZ domain. Kelch-like ECH-associated protein 1 interacts with NF-E2-related factor 2 in a redox-sensitive manner and the dissociation of the proteins in the cytoplasm is followed by transportation of NF-E2-related factor 2 to the nucleus. This interaction results in the expression of the catalytic subunit of gamma-glutamylcysteine synthetase.

Additional Next-Generation Sequencing results continued on the next page. >

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Mutational Analysis by Next-Generation Sequencing (NGS)

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
KRAS	DNA-Tumor	Pathogenic Variant	p.G12C	2	c.34G>T	43	NM_004985.4

Interpretation: A pathogenic codon 12 mutation was detected in KRAS.

KRAS or V-Ki-ras2 Kirsten rat sarcoma viral oncogene homolog encodes a signaling intermediate involved in many signaling cascades including the EGFR pathway. KRAS somatic mutations have been found in pancreatic (57%), colon (35%), lung (16%), biliary tract (28%), and endometrial (15%) cancers. Several germline mutations of KRAS (V14I, T58I, and D153V amino acid substitutions) are associated with Noonan syndrome.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
STK11	DNA-Tumor	Likely Pathogenic Variant	p.P179L	4	c.536C>T	23	NM_000455.4

Interpretation: This STK11 missense mutation is presumed to be pathogenic due to its recurrence in cancers.

STK11 also known as LKB1, is a serine/threonine kinase. It is thought to be a tumor suppressor gene which acts by interacting with p53 and CDC42. It modulates the activity of AMP-activated protein kinase, causes inhibition of mTOR, regulates cell polarity, inhibits the cell cycle, and activates p53. Somatic mutations in this gene are associated with a history of smoking and KRAS mutation in NSCLC patients. The frequency of STK11 mutation in lung adenocarcinomas ranges from 7%-30%. STK11 loss may play a role in development of metastatic disease in lung cancer patients. Mutations of this gene also drive progression of HPV-induced dysplasia to invasive, cervical cancer and hence STK11 status may be exploited clinically to predict the likelihood of disease recurrence. Germline mutations in STK11 are associated with Peutz-Jeghers syndrome which is characterized by early onset hamartomatous gastro-intestinal polyps and increased risk of breast, colon, gastric and ovarian cancer.

Additional Next-Generation Sequencing results continued on the next page. >

PATIENT:

276125

PHYSICIAN: JORDAN MILLER

Mutational Analysis by Next-Generation Sequencing (NGS)

GENES TESTED WITH INDETERMINATE* RESULTS BY TUMOR DNA SEQUENCING					
COL2A1	SMARCA2				

* Genes in this table were ruled indeterminate due to low coverage for some or all exons.

For a complete list of genes tested, visit www.CarisMolecularIntelligence.com/profilemenu.

NGS Methods

Direct sequence analysis was performed on genomic DNA isolated from a micro dissected tumor sample using Illumina NovaSeq 6000 sequencers. A hybrid pull down panel of baits was used to enrich more than 700 clinically relevant genes along with > 20,000 other genes. Sequence data is analyzed using a customized bioinformatics pipeline to detect sequencing variants, copy number alterations (amplifications and deletions) indels and HLA genotypes. In addition, genomic signatures for tumor mutational burden (TMB), microsatellite instability (MSI), genomic loss of heterozygosity (LOH) or HRD Genomic Scar Score (HRD GSS), and homologous recombination deficiency (HRD) are reported when applicable. For a complete list of what is covered by the assay, and genes with partial coverage, please contact Caris Customer Support.

Copy Number Alterations by Next-Generation Sequencing (NGS)

CNA Methods

The copy number alteration (CNA) of each exon is determined by a calculation using the average sequencing depth of the sample along with the sequencing depth of each exon and comparing this calculated result to a pre-calibrated value. A complete list of genes for reporting copy number alterations, including amplifications and deletions, is available upon request.

Gene Fusion and Transcript Variant Detection by RNA Sequencing

GENES TESTED WITH GENE FUSION OR TRANSCRIPT VARIANT DETECTED				
Biomarker	Fusion/Isoform	Splice Site	Transcript ID	Variant Interpretation
MET	SND1:MET	exon 14:exon 6	NM_014390.4/NM_001127500.3	Likely Pathogenic Fusion

Interpretation: A SND1::MET fusion was detected by RNA sequencing. This fusion is predicted to be in frame and encode an intact MET kinase domain. Exon 16 of SND1 (NM_014390.4) is joined in frame to exon 6 of MET (NM_001127500.3).

C-Met is a proto-oncogene that encodes the tyrosine kinase receptor of hepatocyte growth factor (HGF) or scatter factor (SF). c-Met mutation causes aberrant MET signaling in various cancer types including renal papillary, hepatocellular, head and neck squamous, gastric carcinomas and non-small cell lung cancer. Mutations in the juxtamembrane domain (exon 14, 15) results in the constitutive activation and show enhanced tumorigenicity. Germline mutations in c-MET have been associated with hereditary papillary renal cell carcinoma.

Whole Transcriptome Sequencing (WTS) Methods

Gene fusion and variant transcript detection were performed on RNA isolated from a tumor sample using next generation sequencing. The assay also detects fusions occurring at known and novel breakpoints within genes. The genes included in this report represent the subset of genes associated with cancer. The complete list of unclassified alterations is available by request.

Protein Expression by Immunohistochemistry (IHC)

Biomarker	Patient Tumor			Thresholds
	Staining Intensity (0, 1+, 2+, 3+)	Percent of cells	Result	Conditions for a Positive Result:
ALK	0	100	Negative	Intensity $\geq 3+$ and $\geq 1\%$ of cells stained
ERBB2 (Her2/Neu)	0	100	Negative	Intensity $\geq 3+$ and $\geq 10\%$ of cells stained
MET	1 +	20	Negative	Intensity $\geq 3+$ and $\geq 25\%$ of cells stained

PD-L1 TUMOR CELL STAINING				
Biomarker	Patient Tumor			Thresholds
	Staining Intensity (0, 1+, 2+, 3+)	Percent of cells	Result	Conditions for a Positive Result:
PD-L1 (28-8)	1 +	5%	Positive	Intensity $\geq 1+$ and $\geq 1\%$ of cells stained
PD-L1 (SP142)	0	100%	Negative	Intensity $\geq 1+$ and $\geq 50\%$ of cells stained
PD-L1 (SP263)	0	100%	Negative	Intensity $\geq 1+$ and $\geq 1\%$ of cells stained

PD L1 (28 8): Scoring was based on percentage of viable tumor cells showing partial or complete membrane staining at any intensity.

PD L1 (SP142): TC scoring was based on the presence of discernible PD L1 membrane staining of any intensity in $\geq 50\%$ of viable tumor cells.

PD L1 (SP263): Scoring was based on percentage of viable tumor cells showing partial or complete membrane staining at any intensity.

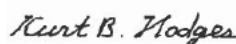
PD-L1 TUMOR PROPORTION SCORE (TPS)			
Biomarker	Result	TPS	Threshold
PD-L1 (22c3)	Positive	1%	TPS $\geq 1\%$

PD L1 22c3: Scoring was based on the percentage of viable tumor cells showing partial or complete membrane staining. There are three categories of PD L1 expression defined by the PD L1 22c3 IHC pharmDx NSCLC interpretation guide: TPS < 1% (negative), TPS $\geq 1\%$ and TPS $\geq 50\%$.

PD-L1 IMMUNE CELL (IC) SCORE			
Biomarker	Result	IC	Threshold
PD-L1 (SP142)	Negative	1%	$\geq 10\%$

PD L1 (SP142): IC scoring was based on discernible PD L1 staining of any intensity in tumor infiltrating immune cells covering $\geq 10\%$ of tumor area occupied by tumor cells, associated intratumoral or contiguous peritumoral stroma.

Clones used: ALK (D5F3), ERBB2 (Her2/Neu) (4B5), MET (SP44) RxDx Assay, PD L1 (22c3), PD L1 (28 8), PD L1 (SP142), PD L1 (SP263).



Electronic Signature
Kurt Hodges, MD, MBA
12/03/2024

Additional IHC results continued on the next page. >

PATIENT

TN24-276125

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Protein Expression by Immunohistochemistry (IHC)

IHC Methods

The Laboratory Developed Tests (LDT) immunohistochemistry (IHC) assays were developed and their performance characteristics determined by Caris Life Sciences. These tests have not been cleared or approved by the US Food and Drug Administration. The FDA has determined that such clearance or approval is not currently necessary. Interpretations of all immunohistochemistry (IHC) assays were performed manually or with the assistance of an AI based image analysis tool by a board certified pathologist using a microscope and/or digital whole slide image(s).

The following IHC assays were performed using FDA approved companion diagnostic or FDA cleared tests consistent with the manufacturer's instructions: ALK (VENTANA ALK (D5F3) CDx Assay, Ventana), ER (CONFIRM anti Estrogen Receptor (ER) (SP1), Ventana), FOLR1 (VENTANA FOLR1 2.1 RxDx, Ventana), CLDN18 (VENTANA, 43 14A RxDx Assay), PR (CONFIRM anti Progesterone Receptor (PR) (1E2), Ventana), HER2/neu (PATHWAY anti HER 2/neu (4B5), Ventana), Ki 67 (MIB 1 pharmDx, Dako), PD L1 22c3 (pharmDx, Dako), PD L1 SP142 (VENTANA, non small cell lung cancer), PD L1 28 8 (pharmDx, Dako, gastric / GEJ, non small cell lung cancer), PD L1 SP263 (Ventana, non small cell lung cancer), and Mismatch Repair (MMR) proteins (MLH1, MSH2, MSH6, and PMS2; VENTANA MMR RxDx Panel, Ventana).

HER2 results and interpretation follow the ASCO/CAP scoring criteria. Bartley, A.N., J.A. Ajani, et al. (2016). "HER2 testing and clinical decision making in gastroesophageal adenocarcinoma: guideline from the College of American Pathologists, American Society for Clinical Pathology, and the American Society of Clinical Oncology". J Clin Oncol. 35(4):446 464.

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#	Drug	Biomarker	Reference
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2	alectinib	ALK	Gadgeel, S., D.R. Camidge, et al. (2018). "Alectinib versus crizotinib in treatment-naïve anaplastic lymphoma kinase-positive (ALK+) non-small-cell lung cancer: CNS efficacy results from the ALEX study." Ann Oncol 29 (11): 2214-2222. View Citation Online
3	alectinib, brigatinib, ceritinib, crizotinib, lorlatinib	ALK	Lindeman, N.I., Y. Yatabe, et al. (2018). "Updated Molecular Testing Guideline for the Selection of Lung Cancer Patients for Treatment With Targeted Tyrosine Kinase Inhibitors Guideline From the College of American Pathologists, the International Association for the Study of Lung Cancer, and the Association for Molecular Pathology." J Thorac Oncol 13(3): 323-358. View Citation Online
4	alectinib, brigatinib, ceritinib, crizotinib, lorlatinib	ALK	National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology. Non-Small Cell Lung Cancer Version 1.2020
5	brigatinib	ALK	Camidge, D.R., S. Popat, et al. (2018). "Brigatinib versus Crizotinib in ALK-Positive Non-Small-Cell Lung Cancer." N Engl J Med 379(21): 2027-2039. View Citation Online
6	brigatinib	ALK	Lin, J.L., G.J. Riely, et al. (2018). "Brigatinib in Patients with Alectinib-refractory ALK-positive NSCLC." J Thorac Onc 13(10): 1530-1538. View Citation Online
7	brigatinib	ALK	Reckamp, K., J. Lee, et al. (2019). "Comparative efficacy of brigatinib versus ceritinib and alectinib in patients with crizotinib-refractory anaplastic lymphoma kinase-positive non-small cell lung cancer." Curr Med Res Opin 35(4):569-576. View Citation Online
8	brigatinib, crizotinib	ALK	Thorne-Nuzzo, T., P. Towne, et al. (2017). "A Sensitive ALK Immunohistochemistry Companion Diagnostic Test Identifies Patients Eligible for Treatment with Crizotinib." J Thorac Oncol 12(5): 804-813. View Citation Online
9	ceritinib, lorlatinib	ALK	Shaw, A.T., E. Felip, et al. (2017). "Ceritinib versus chemotherapy in patients with ALK-rearranged non-small-cell lung cancer previously given chemotherapy and crizotinib (ASCEND-5): a randomised, controlled, open-label, phase 3 trial." Lancet Oncol 18 (7):874-886. View Citation Online
10	ceritinib, lorlatinib	ALK	Solomon, B. J., A. T. Shaw, et al. (2018). "Lorlatinib in patients with ALK-positive non-small cell lung cancer: results from a global phase 2 study." Lancet Oncol 19:1654-1667. View Citation Online
11	ceritinib, lorlatinib	ALK	Soria, J.C., de Castro G., et al. (2017). "First-line ceritinib versus platinum-based chemotherapy in advanced ALK-rearranged non-small-cell lung cancer (ASCEND-4): a randomised, open-label, phase 3 study." Lancet 389: 917-929. View Citation Online
12	crizotinib	ALK	Solomon, B. J., T. S. Mok, et al. (2018). "Final overall survival analysis from a study comparing first-line crizotinib versus chemotherapy in ALK-mutation positive Non-small-cell-lung cancer." J Clin Oncol 36: 2251-2258. View Citation Online
13	crizotinib	ALK	van der Wekken, A. J., H.J.M. Groen, et al. (2017). "Dichotomous ALK IHC is a better predictor for ALK inhibition outcome than traditional ALK FISH in advanced Non-small cell lung cancer." Clin Cancer Res 23(15): 4251-4258. View Citation Online
14	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Hyman, D.H., J. Baselga, et al. (2015). "Vemurafenib in Multiple Nonmelanoma Cancers with BRAF V600 Mutations." NEJM 373(8):726-736. View Citation Online
15	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Planchard, D., B.E. Johnson, et al. (2016). "An open-label phase II trial of dabrafenib (D) in combination with trametinib (T) in patients (pts) with previously treated BRAF V600E mutant advanced non-small cell lung cancer (NSCLC; BR113928)." J Clin Oncol 34: 15_suppl, 107-107. View Citation Online
16	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Planchard, D., B.E. Johnson, et al. (2017). "Dabrafenib plus trametinib in patients with previously untreated BRAFV600E-mutant metastatic non-small-cell lung cancer: an open-label, phase 2 trial." Lancet Oncol 18(1):1307-1316. View Citation Online

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#	Drug	Biomarker	Reference
17	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Riely, G. J., B. E. Johnson, et al. (2023). "Phase II, Open-Label Study of Encorafenib Plus Binimetinib in Patients With BRAFV600-Mutant Metastatic Non-Small-Cell Lung Cancer." <i>J Clin Oncol</i> . 41 (21): 3700-3711. View Citation Online
18	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Subbiah, V., D.M. Hyman, et al. (2019). "Efficacy of Vemurafenib in Patients With Non-Small-Cell Lung Cancer With BRAF V600 Mutation: An Open-Label, Single-Arm Cohort of the Histology-Independent VE-BASKET Study." <i>JCO Precis Oncol</i> 2019; 3, 1-9. View Citation Online
19	afatinib, dacomitinib, osimertinib	EGFR	Sequist, L.V., M. Schuler, et al. (2013). "Phase III Study of Afatinib or Cisplatin Plus Pemetrexed in Patients with Metastatic Lung Adenocarcinoma With EGFR Mutations." <i>J Clin Oncol</i> ahead of print July 1, 2013, doi: 10.1200/JCO.2012.44.2806. View Citation Online
20	afatinib, dacomitinib, osimertinib	EGFR	Wu, Y-L, T.S. Mok, et al. (2017). "Dacomitinib versus gefitinib as first-line treatment for patients with EGFR-mutation-positive non-small-cell lung cancer (ARCHER 1050): a randomized, open-label, phase III trial." <i>Lancet Oncol</i> . 2017;18(11):1454-1466. View Citation Online
21	afatinib, dacomitinib, osimertinib	EGFR	Yang, J.C., et al. (2013). "Symptom control and quality of life in LUX-Lung 3: a phase III study of afatinib or cisplatin/pemetrexed in patients with advanced lung adenocarcinoma with EGFR mutations." <i>J Clin Oncol</i> 31:3342-3350. View Citation Online
22	afatinib, dacomitinib, osimertinib	EGFR	Yang, J.C., et al. (2015). "Clinical activity of afatinib in patients with advanced non-small-cell lung cancer harbouring uncommon EGFR mutations: a combined post-hoc analysis of LUX-Lung 2, LUX-Lung 3, and LUX-Lung 6." <i>Lancet Oncol</i> ; (7):830-838. View Citation Online
23	erlotinib, gefitinib	EGFR	Brugger, W., F. Cappuzzo, et al. (2011). "Prospective molecular marker analyses of EGFR and KRAS from a randomized, placebo-controlled study of erlotinib maintenance therapy in advanced non-small-cell lung cancer." <i>J. Clin. Oncol</i> . 29:4113-4120. View Citation Online
24	erlotinib, gefitinib	EGFR	Fukuoka, M., T.S.K. Mok, et al. (2011). "Biomarker analyses and final overall survival results from a phase III, randomized, open-label, first-line study of gefitinib versus carboplatin/paclitaxel in clinically selected patients with advanced non-small-cell lung cancer in Asia (IPASS)." <i>J. Clin. Oncol</i> . DOI: 10.1200/JCO.2010.33.4235. View Citation Online
25	erlotinib, gefitinib	EGFR	Keedy, V.L., G. Gianconne, et al. (2011). "American Society of Clinical Oncology Provisional Clinical Opinion: epidermal growth factor receptor (EGFR) mutation testing for patients with advanced non-small cell lung cancer considering first-line EGFR tyrosine kinase inhibitor therapy." <i>J. Clin. Oncol</i> . 29(15):2121-2127. View Citation Online
26	erlotinib, gefitinib	EGFR	Maemondo, M., T. Nukiwa, et al. (2010). "Gefitinib or chemotherapy for non-small cell lung cancer with mutated EGFR." <i>N. Engl. J. Med</i> . 362:2380-8. View Citation Online
27	adagrasib, sotorasib	KRAS	Hong, D.S., B.T. Li, et al. (2020). "KRAS G12C Inhibition with Sotorasib in Advanced Solid Tumors." <i>N Engl J Med</i> . 383(13): 1207-1217. View Citation Online
28	adagrasib, sotorasib	KRAS	Janne, P. A., A. I. Spira, et al. (2022). "Adagrasib in Non-Small-Cell Lung Cancer Harboring a KRASG12C Mutation." <i>N Engl J Med</i> . 14;387(2): 120-131. View Citation Online
29	adagrasib, sotorasib	KRAS	Li, B.T, J. Wolf, et al. (2021). "CodeBreak 100: Registrational Phase 2 Trial of Sotorasib in KRAS p.G12C Mutated Non-small Cell Lung Cancer". <i>J Thorac Oncol</i> . 16 (3): S61. View Citation Online
30	capmatinib, tepotinib	MET	Paik, P. K., X. Le, et al. (2020). "Tepotinib in patients with MET exon 14 (METex14) skipping advanced NSCLC: Updated efficacy results from VISION Cohort A." <i>IASLC 2020 World Conference on Lung Cancer</i> (Abstract: 1361) View Citation Online
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37	pembrolizumab	PD-L1 (22c3)	Mok, T.S., KEYNOTE-042 Investigators, et al. (2019). "Pembrolizumab versus chemotherapy for previously untreated, PD-L1-expressing, locally advanced or metastatic non-small-cell lung cancer (KEYNOTE-042): a randomised, open-label, controlled, phase 3 trial." Lancet. 393(10183):1819-1830 View Citation Online
38	pembrolizumab	PD-L1 (22c3)	Reck, M., JR Brahmer, et al. (2019). "Updated Analysis of KEYNOTE-024: Pembrolizumab Versus Platinum-Based Chemotherapy for Advanced Non-Small-Cell Lung Cancer With PD-L1 Tumor Proportion Score of 50% or Greater." J Clin Oncol. 37(7):537-546 View Citation Online
39	nivolumab/ipilimumab combination	PD-L1 (28-8)	Hellmann, M.D., S.S. Ramalingam, et al., (2019). "Nivolumab plus Ipilimumab in Advanced Non-Small-Cell Lung Cancer." N Engl J Med 381:2020-31. View Citation Online
40	pralsetinib	RET	Gainor J.F., V. Subbiah, et al. (2020). "Registrational dataset from the phase I/II ARROW trial of pralsetinib (BLU-667) in patients (pts) with advanced RET fusion+ non-small cell lung cancer (NSCLC)." J Clin Oncol. 38(suppl):9515. View Citation Online
41	pralsetinib, selpercatinib	RET	National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology. Non-Small Cell Lung Cancer Version 6.2020 View Citation Online
42	selpercatinib	RET	Drilon, A., V. Subbiah, et al. (2020). "Efficacy of Selpercatinib in RET Fusion-Positive Non-Small-Cell Lung Cancer." N Engl J Med. 383 (9): 813-824. View Citation Online
43	ceritinib, lorlatinib	ROS1	Lim SM, BC Cho, et al. (2017). "Open-Label, Multicenter, Phase II Study of Ceritinib in Patients With Non-Small-Cell Lung Cancer Harboring ROS1 Rearrangement". J Clin Oncol. May 18;JCO2016713701. View Citation Online
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#	Drug	Biomarker	Reference
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